

EMILIO BERTI

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SUMMARY

I am a theoretical ecologist with a passion for math and computing. I have a strong quantitative background, with expertise in mathematical and statistical modelling of complex systems using big databases at large spatio-temporal scales. I have worked in many different fields of biology, from molecular muscle physiology to macroecology and biogeography. I am very open minded to other cultures and societies: I am Italian, my wife is Bulgarian, we met in Denmark, and we live in Germany. I am an avid reader of ancient history and I love sumo.

PROFESSIONAL EXPERIENCE

Technician

October 2026 – Present

Theory in Biodiversity Science,
German Centre for Integrative Biodiversity Research Halle-Jena-Leipzig – iDiv e.V.,
Leipzig, Germany.

PostDoctoral researcher

October 2020 – 2026

Theory in Biodiversity Science,
German Centre for Integrative Biodiversity Research Halle-Jena-Leipzig – iDiv e.V.,
Leipzig, Germany.

Scientific consultant

May 2020 – July 2020

Department of Bioscience,
Aarhus University,
Aarhus, Denmark.

Teaching assistant

February 2017 – April 2020

Department of Biology,
Aarhus University,
Aarhus, Denmark.

EDUCATION

PhD

February 2017 – June 2020

Section of Ecoinformatics and Biodiversity, Department of Biology, Aarhus University,
Aarhus, Denmark.

Visiting PhD student

Fall 2018

Department of Ecology and Evolution, University of Chicago,
Chicago, IL, USA.

MSc in Biology

2013 – 2016

Department of Ecology and Evolution, University of Florence,
Florence, Italy.

BSc in Biology

2009 – 2012

Department of Physiology, University of Florence,
Florence, Italy.

GRANTS

2024 U. S. National Science Foundation (NSF). Understanding impacts of climatic variability on distributions of species. 800,000\$ awarded to Prof. Daniel Reuman (PI), Emilio Berti (co-PI), Townsend Peterson (co-PI), and Jorge L. Soberon (co-PI).

RELEVANT SKILLS

I have developed an outstanding mathematical and theoretical set of skills and successfully applied it to investigate macroecological and biogeographical drivers of biodiversity.

Programming languages – R, python, bash, C/C++, Stan, javascript (mostly for Google Earth Engine), SQL (postgis flavor).

Software – Linux/GNU, Anaconda, RShiny, Google Earth Engine, High-performance computing (HPC, Slurm flavor), Tidyverse, Git, GitHub, Jupyter Notebooks, L^AT_EX.

Methods – Mathematical modeling, Geographic information systems (GIS), Geoinformatics, Geomatics, Remote sensing, Biostatistics, Data science, Climate analyses, Species distribution modeling, Environmental niche modeling, Machine learning, Bayesian statistics, Ordination and classification, Network theory, Community assembly, Optimization, Automation.

Languages – Italian (native), English (fluent), German (A1 → A2.1).

SOFTWARE

GHCNr: Download and process daily weather data from the Global Historical Climatology Network (GHCN) database. Author, maintainer. <https://cran.r-project.org/packages=GHCNr>.

enerscape: Compute Energy Landscapes (CRAN). Author, maintainer. <https://cran.r-project.org/package=enerscape>.

ATNr: Run Allometric Trophic Networks Models (CRAN). Author. <https://cran.r-project.org/package=ATNr>.

squirrygis: Fast C++ routines to process climate layers (GitHub). Author, maintainer. <https://github.com/emilio-berti/squirrygis>. assembly: Simulate community assembly (GitHub). Author, maintainer. <https://github.com/emilio-berti/assembly>.

SELECTED PUBLICATIONS

A full list of my publications can be found at [Google Scholar](#).

1. **Berti, E.** & Vollrath F. (2026). Energy costs of Hannibal's alpine crossing. *PNAS*. DOI: [10.1073/pnas.2612764123](https://doi.org/10.1073/pnas.2612764123).
2. **Berti, E.**, Rosenbaum, B., & Vollrath F. (2025). Energy landscapes direct the movement preferences of elephants. *Journal of Animal Ecology*. DOI: [10.1111/1365-2656.70023](https://doi.org/10.1111/1365-2656.70023).
3. Li, J., Brose, U., Rosenbaum, B., Ryser, R., & **Berti, E.** (2024). Decoding Information Flow and Sensory Pollution: A Systematic Framework for Understanding Species Interactions. *Ecology Letters*. DOI: [10.1111/ele.14522](https://doi.org/10.1111/ele.14522).
4. Antunes, A. C., **Berti, E.**, ..., & Gauzens, B. (2024). Linking biodiversity, ecosystem function, and Nature's contributions to people: a macroecological energy flux perspective. *Trends in Ecology & Evolution*. DOI: [10.1016/j.tree.2024.01.004](https://doi.org/10.1016/j.tree.2024.01.004).
5. Gauzens, B., Brose, U., Delmas, E., & **Berti, E.** (2023). ATNr: Allometric Trophic Network models in R. *Methods in Ecology and Evolution*.
6. Bauer, B., **Berti, E.**, ... & Brose, U. (2022). Biotic filtering by species' interactions constrains food-web variability across spatial and abiotic gradients. *Ecology letters*. DOI: [10.1111/ele.13995](https://doi.org/10.1111/ele.13995). (Shared first authorship).
7. Grenié, M., **Berti, E.**, ... & Marten, W. (2022). Harmonizing taxon names in biodiversity data: a review of tools, databases, and best practices. *Methods in Ecology and Evolution*. DOI: [10.1111/2041-210X.13802](https://doi.org/10.1111/2041-210X.13802).

8. **Berti, E.**, Davoli, M., ... & Vollrath, F. (2021). The r package enerscape: A general energy landscape framework for terrestrial movement ecology. *Methods in Ecology and Evolution*. DOI: [10.1111/2041-210X.13734](https://doi.org/10.1111/2041-210X.13734).
9. **Berti, E.**, Monsarrat, S., Munk, M., Jarvie, S. & Svenning, J.C. (2020). Body size is a good proxy for vertebrate charisma. *Biological Conservation*. DOI: [10.1016/j.biocon.2020.108790](https://doi.org/10.1016/j.biocon.2020.108790).
10. **Berti, E.** & Svenning, J.C. (2020). Megafauna extinctions have reduced biotic connectivity worldwide. *Global Ecology and Biogeography*. DOI: [10.1111/geb.13182](https://doi.org/10.1111/geb.13182).

CONFERENCE TALKS

- Submitted **Berti, E.**, Rosenbaum, B., Brose, U., & Vollrath, F. (2023). Energy landscapes direct the movement preferences of elephants. *British Ecological Society annual meeting, Belfast, UK*.
- Invited Bauer, B., **Berti, E.**, ..., & Brose, U. (2022). From regional to local scale: biotic interactions shape multilayer food-webs. *SFE-GFO-EEF biannual meeting, Metz, France*.
- Submitted **Berti, E.**, & Svenning, J.C. (2022). State-space models show that functional replacements of extinct megafauna have distinct habitat preference in a European rewilding area. *SFE-GFO-EEF biannual meeting, Metz, France*.
- Submitted Grenié, M., **Berti, E.**, Carvajal-Quintero, J., Winter, M., & Sagouis (2021). Matching Species Names Across Biodiversity Databases: Sources, tools, pitfalls and best practices for taxonomic harmonization. *TDWG annual meeting, online*.
- Submitted **Berti, E.** & Svenning, J.C. (2019). Megalinks extinction and the decrease of ecosystem connectivity. *ESA annual meeting, Louisville, KY*.
- Submitted **Berti, E.**, Jarvie, S. W., & Svenning, J.C. (2018). Rewiring food webs via trophic rewilding. *BES annual meeting, Belfast, UK*.

PEER REVIEW

As of July 2022, I have reviewed 7 papers for: *Ecography* (1), *Ecology Letters* (2), *Ecology* (1), *GigaScience* (1), *Methods in Ecology and Evolution* (1), and *Scientia Agricola* (1). You can find more at my [Publons profile](#).

SUPERVISION, MENTORING, AND TEACHING

I am formally co-supervising the PhD candidate Jingyi Li, who is developing a novel mathematical approach for functional responses based on information theory. In addition, I provide theoretical, computational, and statistical advice to several members of the TiBS working group at iDiv as well as individual mentoring for PhD students, advising especially on transferable skills and alternative career paths outside academia. During my PhD, I taught one course per semester, as part of my salary was paid by Aarhus University on the basis of teaching. In addition, I have regularly organized workshops to teach reproducible research and open data principles and have organized a weekly journal club during my PhD and currently during my PostDoc.

- Meta-analyses for Biodiversity (2024 & 2021) – Teaching assistant (MSc course).
- Theoretical Population Ecology (2023) – Teaching assistant (MSc course).
- Introduction to scientific programming and tidyverse (2022) (slides) – Lecturer (PhD course).
- Introduction to git and GitHub for a fool-proof programming (2022) (course) – Lecturer (PhD course).
- Statistical and Geospatial Modelling (2019) – Teaching assistant (MSc course).
- Behavioural Biology (2018, 2019) – Teaching assistant (BSc course).
- Geographic Information System (2017) – Lab assistant (BSc course).
- Cleaning online repository data for use in biogeography and macroecology (2019) – Co-organizer (workshop).

- Running a species distribution model in R (2019) – Co-organizer (workshop).
- A (very) gentle introduction to Linux (2019) – Organizer (workshop).

COLLABORATIONS

I have moved several times to pursue my career dreams and, being a friendly person, I established personal and professional ties with the people I met. My network of current collaborators include:

- Prof. Jens-Christian Svenning, Aarhus University, Aarhus, Denmark.
- Prof. Ulrich Brose, Jena University, Jena, Germany.
- Prof. Daniel Reuman, Kansas University, Lawrence, KS, USA.
- Prof. Giacomo Santini, Università degli Studi di Firenze, Florence, Italy.
- Prof. Kai Yue, Fujian Normal University, Fuzhou, China.
- Prof. Neil Carter, University of Michigan, Ann Arbor, MI, USA.
- Prof. Fritz Vollrath, University of Oxford, Oxford, UK.